

GenFlex

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Chapter 1

Introduction

This project introduces **GenFlex**, a platform designed to increase the effectiveness and efficiency of examination and evaluation process of educational institutes. It addresses common issues including how long it takes to produce exams, generate lectures, how unevenly grades assignments and how long it takes to give feedback to students through evaluation. As education increasingly shifts to online learning, traditional exam approaches are falling behind. Oftentimes, teachers spend too long to prepare and mark exams by hand, which raises the possibility of errors and delays student response.

System Background:

GenFlex is a system that will be automating the complete exam process, from creation to grading of exams. Examinations may be put up fast by teachers, taken digitally by students, and prompt, reliable evaluation is provided via the platform. This system is adaptable enough to accommodate a range of educational demands. [?].

1.1 Problem Statement

A concise description of the problem:

1. The manual procedures used in the exam systems now in use can cause them to be slower and more error-prone.
2. Exam creation, administration, and grading can take a teacher a great deal of time, detracting from their ability to instruct and engage with students.
3. Because manual evaluation has inherent problems, grading by hand can occasionally result in inconsistent results.
4. Students learn more slowly when grading and comments are delayed, and they wait too long to find out how they did.

5. It is more difficult to evaluate pupils successfully across disciplines when there isn't a single system for managing various question formats, such as multiple-choice, essays, and coding related questions etc.
6. Exam techniques that are out of date find it difficult to meet the demands of increased speed and scalability as education becomes increasingly digitally integrated.
7. Teachers would have less tasks to complete if there was an automated system that was more effective at designing, assessing, and marking tests.
8. A greater variety of question forms, including text, multiple-choice, and diagram based questions, are made possible by digital solutions. Additionally, they guarantee quicker, more equitable grading and quicker student feedback.

1.2 Scope

The goal of **GenFlex** is to give educational institutions a digital platform that completely automates the examination process. With the use of AI technology, the platform would enable teachers to easily create and assess exams. The features listed below will be supported by **GenFlex**:

1. Making of lecture notes and exam papers digitally.
2. Digital exams that students can take via web interface.
3. Text/Descriptive, multiple-choice, code, and math question's grading that is automated utilizing sophisticated AI models like LLMs including coding libraries, math models.
4. Multiple user access: Safe and scalable infrastructure that manages numerous exams which at once will be available to teachers and students alike.

The project's main goal is to automate exams for educational institutions so that they can use a seamless digital platform devoid of the effect of human grading. Teachers may focus on teaching as **GenFlex** ensures fair, accurate, and consistent grading in addition to saving time.

1.3 Problem Solution

GenFlex offers a comprehensive solution to the problem by allowing teachers to easily generate exams and lecture notes.

- Students can take tests/Exams digitally and the system automatically evaluates their submitted responses using some complex AI models such as LLMs along with Coding, and math libraries.
- This digitalization and automation lowers errors and subjectivity in grading and gives immediate feedback by enhancing the whole examination process.

Key objectives include:

1. Automating exam creation and evaluation: Give teachers the ability to swiftly produce exams using a comprehensive question bank and customizable templates. Exams are auto evaluated utilizing AI-driven algorithms, which deliver precise and rapid grading across a number of subjects.
2. Providing a consistent, unbiased grading mechanism: Advanced machine learning algorithms evaluate responses consistently across multiple students, delivering fair outcomes regardless of student identification, background, or subjective teacher interpretation.
3. Supporting several kinds of questions: In addition to text-based questions, the platform also handles coding and mathematical kind of questions.
4. Easing the administrative load on instructors: Our system GenFlex drastically cuts down on the amount of time teachers spend on administrative work by automating exam creation, delivery, grading, and feedback.

1.4 Modules

The exam process is handled by a number of essential modules that make up the GenFlex platform. Every module will possess a unique set of capabilities that enhance the system's overall performance.

1.4.1 Module 1: Teacher Module

This module allows teachers to create exams and lecture notes.

Features:

1. Generate multiple types of questions.

Sub features

- (a) MCQs
 - (b) Essay-type
 - (c) Coding related Feature
2. Upload material like pdf, slides, web links etc to create
- (a) Course notes
 - (b) Exam bank/paper

1.4.2 Module 2: Student Module

This module enables students to attempt exams digitally.

Feature 1:

- Take exams using a web interface.

Feature 2:

- Receive Quick grading/marks after attempting exams.

1.4.3 Module 3: AI Evaluation Module

This module automates the grading process using AI models.

Feature 1:

- Use LLMs for evaluating long-answer questions. Feature

Feature 2:

- Use Math/Coding libraries to deal with coding and math questions.

1.4.4 Module 4: Security and Authentication Module

This module handles the security aspects of the system.

Feature 1:

- User authentication through secure login.

Feature 2:

- Data encryption and protection during exam transmission.

1.5 User Classes and Characteristics

User class	Description
Teacher	Teachers have a tough job when it comes to creating exams, lecture notes from course materials like PDFs, slides, and te other texts. With the GenFlex platform, they can make things easier by automating how exams are made and graded. Most teachers won't need much training to get the expertise of the AI features in our system. They can also take a look at the automated evaluations of the system and adjust the student's grades accordingly if they feel it's necessary to avoid biasness in system.
Student	Students will use the GenFlex platform to give their exams online. After they submit their answers, they'll get their grades back quickly and accurately. Since many students might have the different levels of comfort with the technology and it's important that the platform is simple and easy to use for all of them. It should support various kinds of the questions like they can be multiple-choice questions (MCQ's), descriptive/Essay type questions, coding related or mathematical questions.
IT Staff	The IT Support team helps keep GenFlex running smoothly through out the exam sessions. They take care of the system's security and manage the database along with the fixing any technical issues and problems that come up while the exam hours of the students. During exam times, they make sure everything works very well and without the delays. They also update the platform when needed to improve how it works. Other key part of the IT Staff is to make sure that the users can log into the system securely and that all exam data is kept safe.

Chapter 2

Project Requirements

2.1 Use-cases

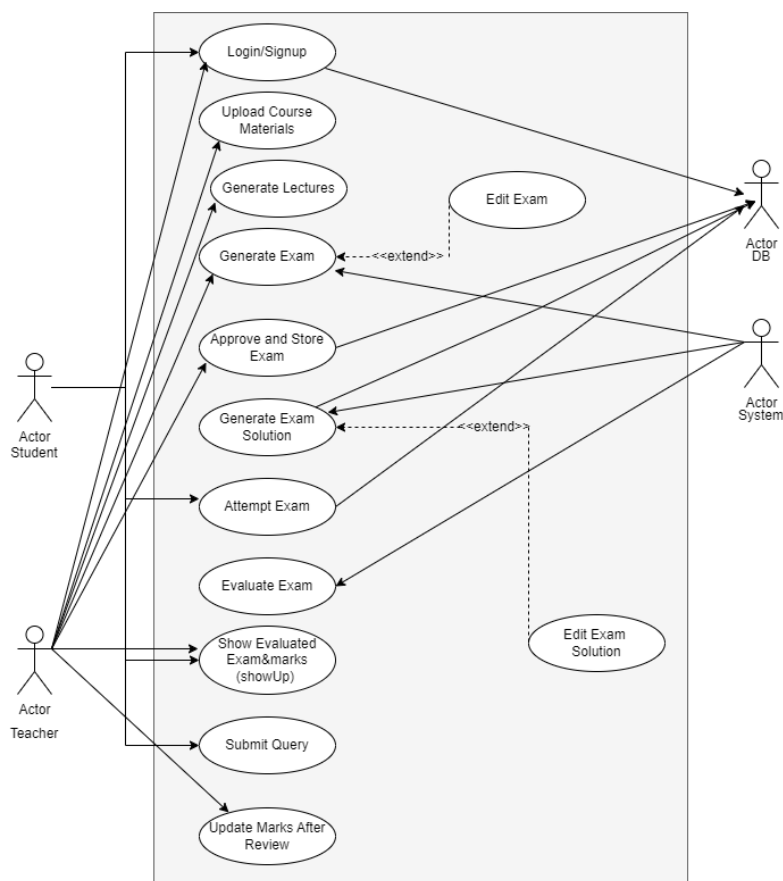


Figure 2.1: Usecase Diagram

2.1.1 Use-case 1

ID: UC01

Name: Login/Signup

Actor: Teacher, Student

Description: Users can log in or sign up to the application based on their roles (faculty or student)

2.1.2 Use-case 2

ID: UC02

Name: Upload Course Materials

Actor: Teacher

Description: Teachers can upload course materials (slides, notes , pdfs etc) to the platform.

2.1.3 Use-case 3

ID: UC03

Name: Generate Exam

Actor: Teacher

Description: System generates an exam based on the uploaded course materials, and the teacher can modify it using an editor.

2.1.4 Use-case 4

ID: UC04

Name: Generate Exam Solution

Actor: Teacher, System

Description: System generates a solution for the exam, and the teacher can edit it before finalizing.

2.1.5 Use-case 5

ID: UC05

Name: Approve and Store Exam

Actor: Teacher

Description: Once the exam and solution are ready, the teacher approves them, and both are stored in the local database.

2.1.6 Use-case 6

ID: UC06

Name: Attempt Exam

Actor: Student

Description: Students log in and attempt the exam. Their answers are stored in the local database.

2.1.7 Use-case 7

ID: UC07

Name: Evaluate Exam

Actor: System, Teacher

Description: The system compares the student's solution with the provided solution and generates marks, which the teacher can modify if necessary.

2.1.8 Use-case 8

ID: UC08

Name: Show Evaluated Exam

Actor: System, Student

Description: The system displays the evaluated exam with detailed marks for each question to students.

2.1.9 Use-case 9

ID: UC09

Name: Submit Query

Actor: Student

Description: Students can submit queries about their marks, and teachers can review them.

2.1.10 Use-case 10

ID: UC10

Name: Update Marks

Actor: Teacher

Description: After reviewing student queries, the teacher can update the marks if required.

2.2 Userstories

1. As a Student, I want to log in/sign up, so I can take exams.
2. As a Teacher, I want to upload course materials, so I can create lectures and exams based on them.
3. As a Teacher, I want the system to generate an exam, so I can review and modify it.

Sub-story:

As a Teacher, I want to modify the exam, so I can make it better.

4. As a Teacher, I want the system to generate a solution so that I can have a reference for evaluation after taking exams.

Sub-story:

As a Teacher, I want to edit the solution so that I can ensure the solution is correctly generated.

5. As a Teacher, I want to approve and store the exam, so it is saved in the system for future use.
6. As a Student, I want to attempt an exam, so I can complete the assigned exam.

7. As a System, I want to evaluate the student's answers, so I can automatically provide marks

Sub-story:

As a Teacher, I want to modify the evaluations done by the system, so I can ensure it is correct.

8. As a Teacher, I want to show the evaluated exam to the student, so they can see their marks.
9. As a Student, I want to submit a query so that I can get clarification if my marks are correctly checked.
10. As a Teacher, I want to review student queries, so I can address their concerns.

Sub-story:

As a Teacher, I want to update marks after reviewing the student queries so that I ensure fair evaluation among students.

2.3 Functional Requirements

2.3.1 Module 1: Teacher Module

Following are the requirements for module 1:

1. The system shall allow teachers to create multiple-choice questions (MCQs).
2. The system shall enable teachers to create essay-type questions.
3. The system shall provide functionality for teachers to make coding-related questions.
4. The system shall allow teachers to upload PDF documents for course materials.
5. The system shall enable teachers to upload presentation slides for Lecture notes.
6. The system shall allow teachers to create and manage an exam question bank.
7. The system shall provide facilities for teachers to generate lecture notes from uploaded materials.
8. The system shall offer a user friendly interface for teachers to create exams.
9. The system shall allow teachers to compile exam papers from the question bank.

2.3.2 Module 2: Student Module

Following are the requirements for module 2:

1. The system shall provide a web interface for students to attempt exams digitally.
2. The system shall support various question Displays Like MCQs, essay-type, and coding questions in Exam.
3. The system shall provide quick grading/marks to students after they attempt exams.
4. The system shall display detailed feedback for each question type to students.

2.3.3 Module 3: AI Evaluation Module

Following are the requirements for module 3:

1. The system shall use Large Language Models (LLMs) to evaluate descriptive or essay-type questions after submission by students.
2. The system shall utilize coding libraries to evaluate coding-related question answers submitted by students.
3. The system shall support programming languages for coding evaluations submitted by students.
4. The system shall use math libraries to deal with mathematical questions.

2.3.4 Module 4: Security and Authentication Module

Following are the requirements for module 4:

1. The system shall provide secure login functionality for user authentication to make sure that only right person have access.
2. The system shall implement role-based access control to make sure to show teachers and students their corresponding screens.
3. The system shall implement encryption for exam data during transmission when submitted by students.
4. The system shall ensure the security of stored exam questions and student responses on database to avoid any kind of loss of data.

5. The system shall ensure and strengthen the database security as it contains sensitive information e.g exam questions, student personal information, and evaluation results of students.

2.4 Non-Functional Requirements

2.4.1 Reliability

1. The GenFlex system shall have a Mean Time Between Failures (MTBF) of at least 168 hours (7 days) of continuous operation during the initial prototype phase.
2. A failure is defined as any unplanned system downtime or inability to access critical functionalities (exam creation, taking, or result viewing) for more than 10 minutes.
3. In the failure case, the system will automatically save all user progress made in the last 2 minutes to ensure that current ongoing exams are not disturbed and reduce any data loss.
4. The system will keep track of the errors and performance problems and then allow them to be manually reviewed and checked accordingly. Automated detection and alerts may be added in upcoming versions.
5. If a failure occurs, the system will automatically recover and restore services within 30 minutes. Manual intervention will only be needed when needed urgently.

2.4.2 Usability

1. The GenFlex system shall allow teachers to create a standard exam with at least 20 questions within 60 minutes on their first use.
2. Students shall be able to navigate and start an assigned exam within 5 clicks from the main dashboard on Student tab/window.
3. The system shall provide basic error messages(error prompt messages) if user put wrong inputs.
4. The system will let users easily undo their last action in form fields, either by using familiar keyboard shortcuts (ctrl+z or a convenient "undo" button), wherever possible.

2.4.3 Performance

1. The system shall support up to 100 concurrent users simultaneously during the initial phase, with a goal of expanding its limit upto 1000 users as the system is optimized.
2. 80 percent of all database queries, including those for retrieving exam questions and saving responses, shall complete in less than 1 second.
3. The AI evaluation module shall complete the grading of a standard 50 questions with mixedup format exam (MCQs and short answers only) within 5 minutes.
4. The system shall render the user interface for exam-taking within 5 seconds of a page request under normal load conditions.
5. Auto-saving of responses during exams shall occur every 60 seconds, ensuring user interaction is not interrupted if any kind of failures occur like electricity outage etc.

2.4.4 Security

1. The system shall use TLS 1.2 or might be a higher one to keep all data secure and safe during transmission, and sensitive data stored in the system shall be protected by AES-128 encryption at first, with plans to upgrade to AES-256 in future versions.
2. Teachers and administrators shall use single-factor authentication (like a password) in initial release, and multi-factor authentication shall be added later in later updates.
3. The system shall be designed to handle common security issues, following the OWASP Top 10 instructions,.
4. All user actions shall be held for the audits and the checkups to keep the track of activities done while interacting with the system throughout the session.
5. The system shall lock accounts of the user if 10 failed login attempts occur within the 30 minutes time frame, along with the option for the users to recover their account through some email-based process.
6. The system shall make sure the password policies make strong passwords, with a requirement of having the minimum of 12 characters, including uppercase Alphabets, lowercase Alphabets, numerics, and special characters, with mandatory changes every 90 days.

2.5 Domain Model

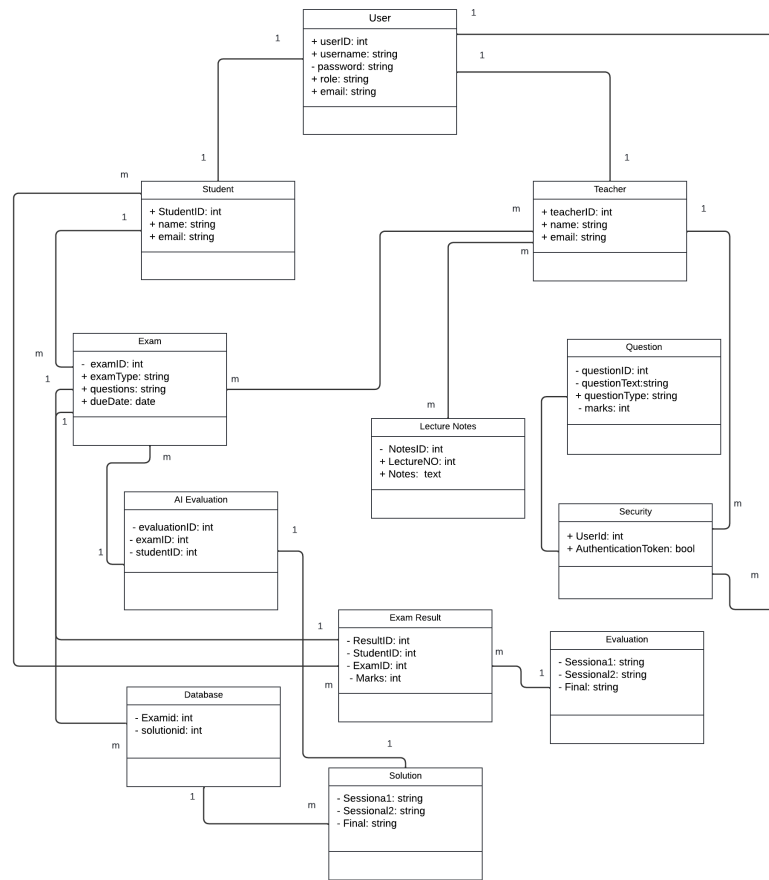


Figure 2.2: Domain Model Diagram

2.6 Context Diagram

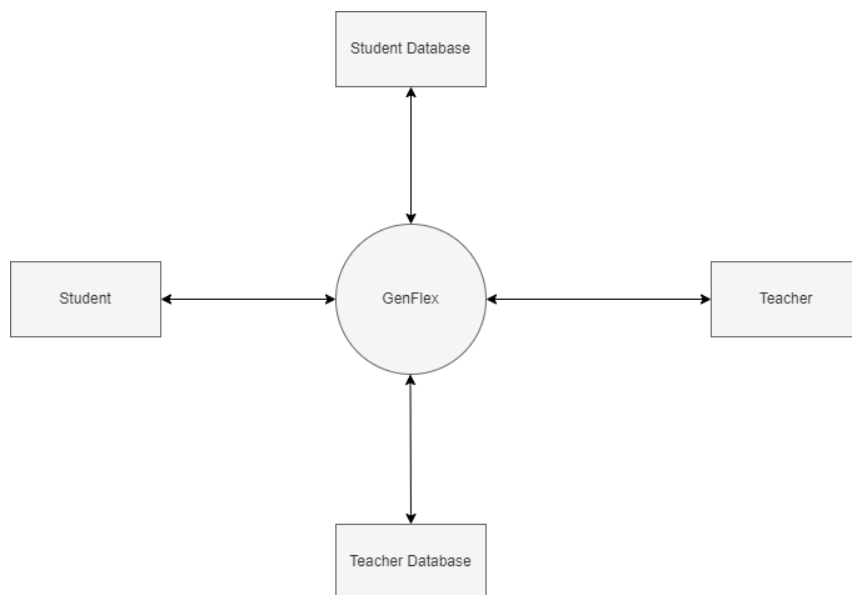


Figure 2.3: Context Diagram

Chapter 3

System Overview

Give a general description of the functionality, context, and design of your project. Provide any background information if necessary.

3.1 Architectural Design

GenFlex uses a layered architecture with five main layers:

1. Presentation Layer: Web and mobile interfaces
2. Application Layer: Teacher, Student, and Admin modules
3. Service Layer: AI Evaluation
4. Data Access Layer: Database and file system operations
5. Data Layer: MongoDB and file storage

Key modules include:

- Teacher Module: Exam creation and management
- Student Module: Exam taking and result viewing
- AI Evaluation Module: Automated grading using LLMs and specialized libraries
- Exam Management: Handles exam lifecycle
- Security and Authentication: Ensures system-wide data protection

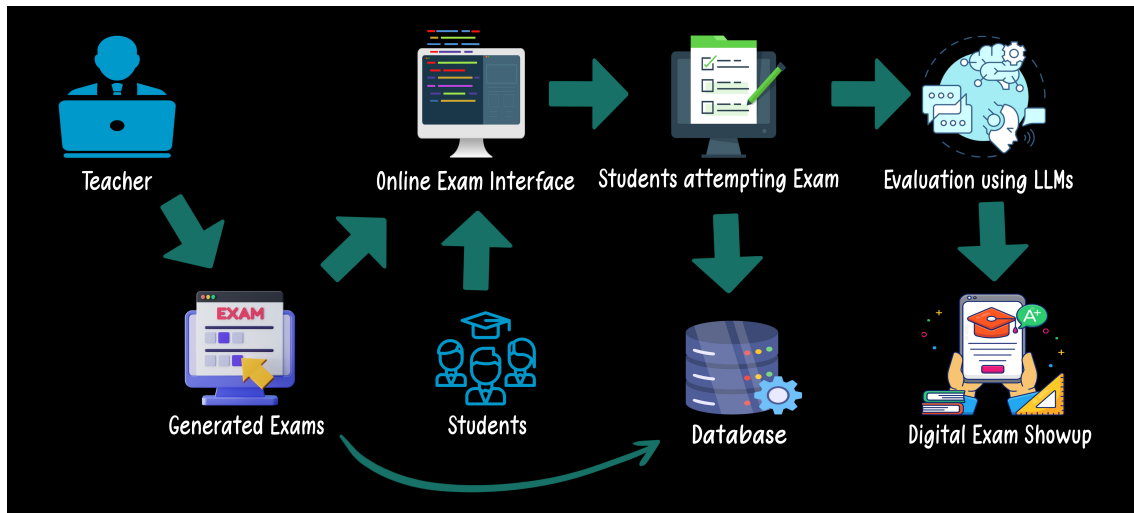


Figure 3.1: Architecture Diagram

The architecture allows for separation of concerns, scalability, and easy integration of new features. Modules collaborate through defined interfaces, with data flowing from the presentation layer through the application and service layers to the data layer and back, enabling the full exam automation process.

3.2 Design Models

Design model applicable our system is Object Oriented Development Approach.

Design Models for Object Oriented Development Approach

The applicable models for the project using object oriented development approach may include:

- Activity Diagram
- Class Diagram
- Class-level Sequence Diagram
- State Transition Diagram (for the projects which include event handling and backend processes)

3.2.1 Teacher Activity Diagram

Description:

This activity diagram outlines the workflow for a teacher using the GenFlex system. It starts with the teacher logging in and progresses through course material upload, exam

generation, and evaluation processes. The diagram includes decision points for editing exams and solutions, as well as for modifying AI-generated evaluations. It concludes with handling student queries.

Usage:

This diagram guides the development of the teacher interface and workflow in GenFlex. It ensures that all necessary steps are included, from login to query resolution.

- Secure login system
- Course material upload and lecture generation
- Exam creation and editing tools
- AI-driven exam evaluation
- Manual evaluation adjustment
- Student result publication
- Query handling system

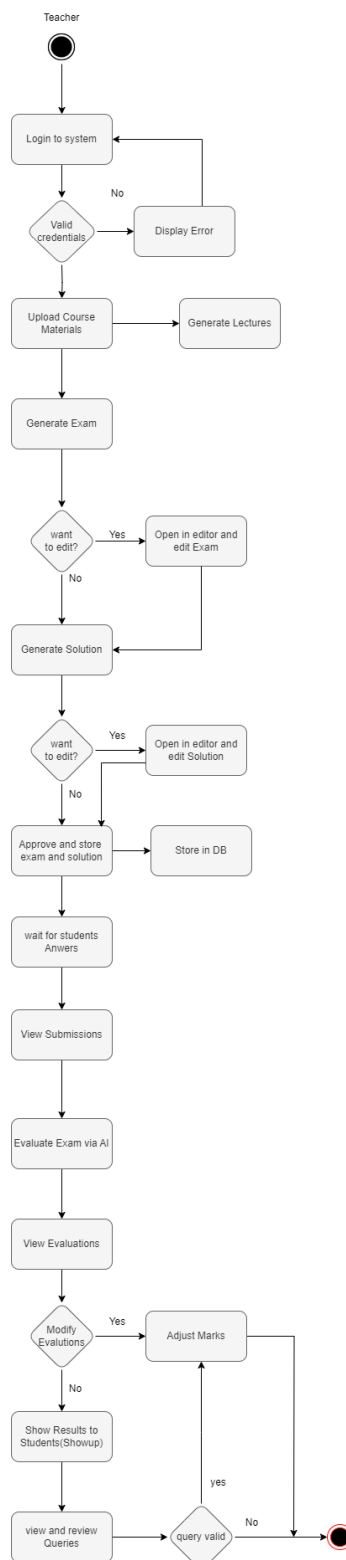


Figure 3.2: Teacher Activity Diagram

3.2.2 Student Activity Diagram

Description:

This activity diagram illustrates the student's journey in the GenFlex system. It begins with the student logging in and proceeds through viewing available exams, attempting an exam, and submitting it. The diagram also covers the evaluation viewing process and includes a query submission feature with a validation step.

Usage:

This diagram serves as a blueprint for developing the student interface and experience in GenFlex. It ensures a logical flow from exam access to result viewing. Developers can use this to implement:

- Secure student authentication
- Exam viewing and attempt interface
- Exam submission and storage system
- Result viewing mechanism
- Query submission and handling process

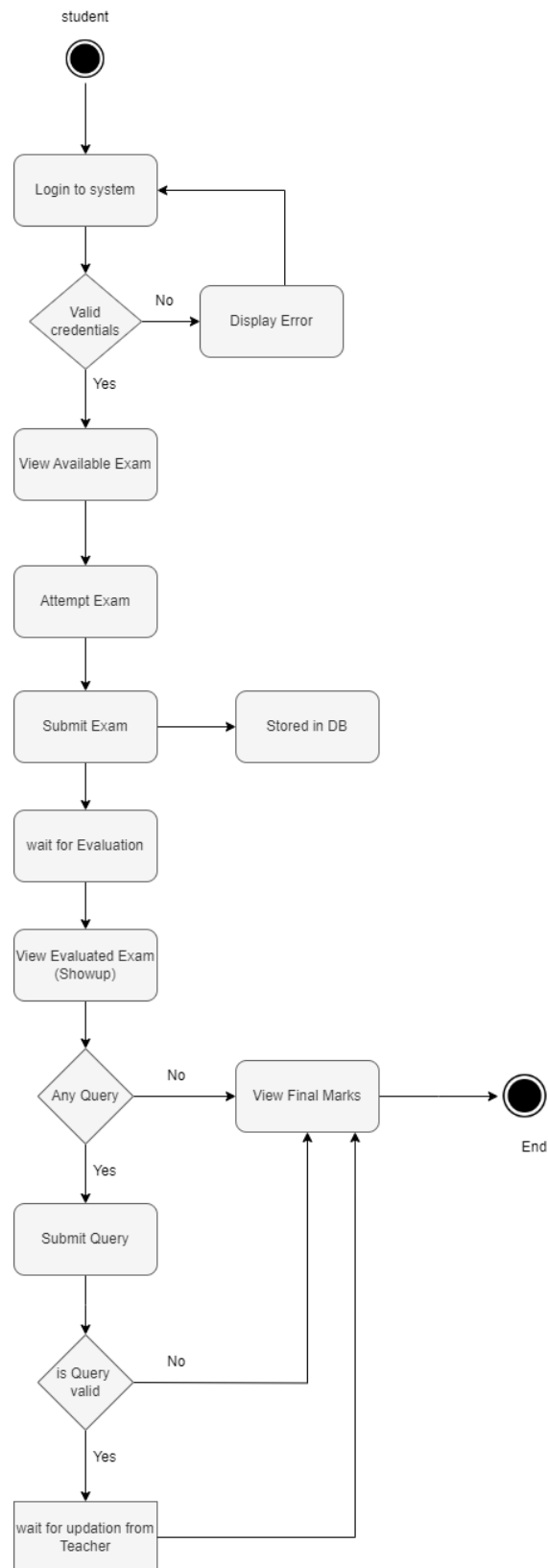


Figure 3.3: Students Activity Diagram

3.2.3 Class Diagram

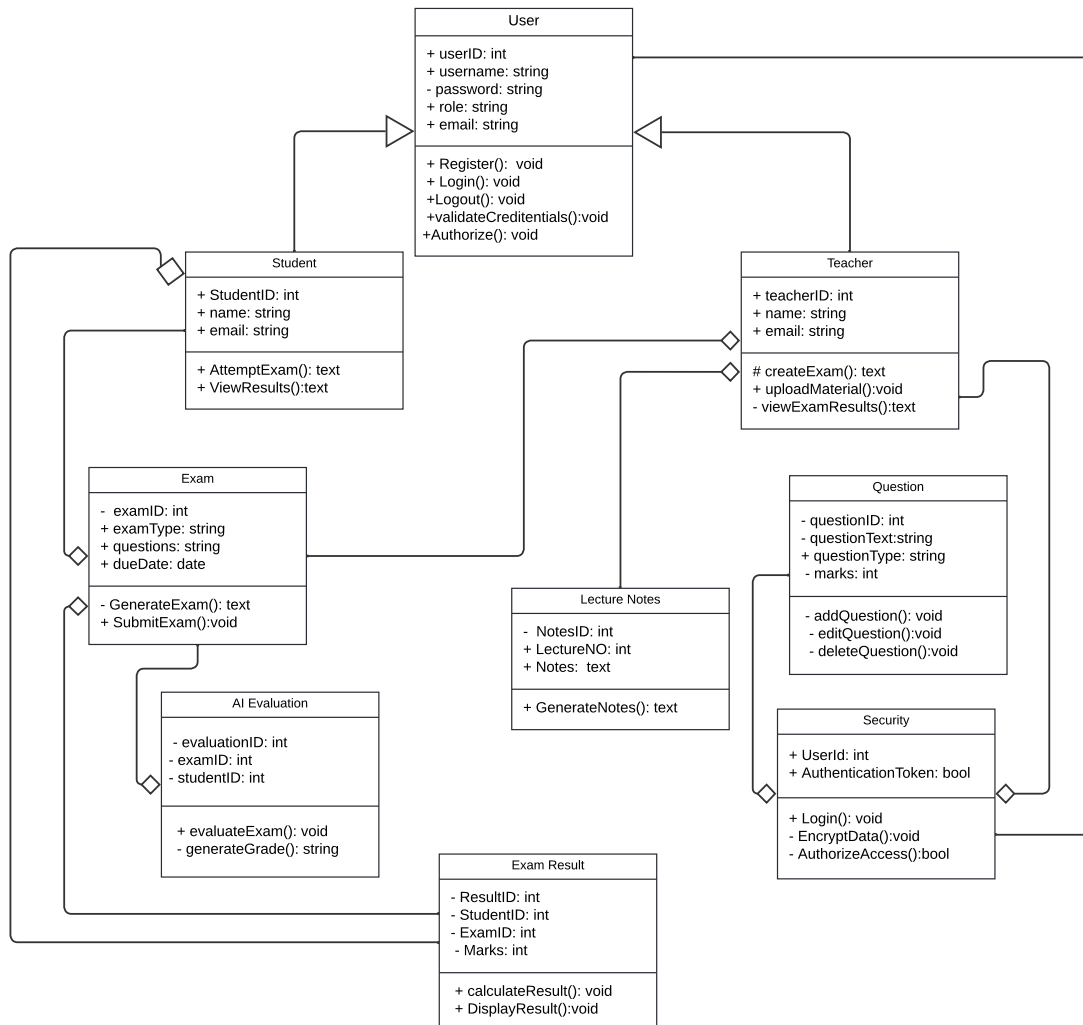


Figure 3.4: Class Diagram

3.2.4 Teacher Exam Creation Sequence Diagram

Description:

This outlines the process of a teacher creating an exam in GenFlex. It illustrates the sequence from the teacher initiating exam creation through the User Interface, to the Exam Controller managing the creation process. The diagram shows how the system generates questions using a Question Generator, saves the exam data in the Database, and confirms the creation to the teacher. This process demonstrates the automated exam creation feature of GenFlex.

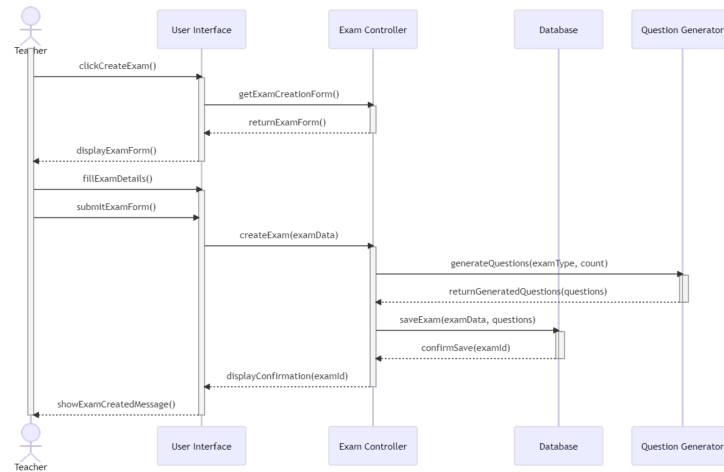


Figure 3.5: Exam Creation Sequence Diagram

3.2.5 Student Exam-Taking Sequence Diagram

Description:

This diagram depicts the sequence of interactions when a student takes an exam in Gen-Flex. It shows the flow from the student initiating the exam through the User Interface, to the Exam Controller fetching exam details from the Database, rendering the exam, processing the student's submitted answers, and finally using the AI Evaluator to grade the exam. The process concludes with saving the results and displaying them to the student.

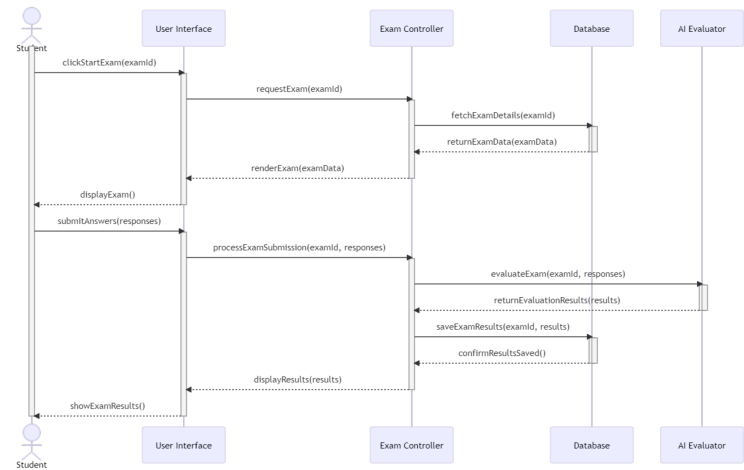


Figure 3.6: Student Exam-Taking Sequence Diagram

3.2.6 AI Evaluation Sequence Diagram

Description:

This diagram illustrates the AI-powered evaluation process in GenFlex. It shows how the Exam Controller initiates the evaluation, and how the AI Evaluator processes different types of questions (essay, coding, and math) using specialized components like the Language Model, Coding Library, and Math Library. The process includes categorizing questions, evaluating each type separately, and aggregating results before returning the final evaluation to the Exam Controller.

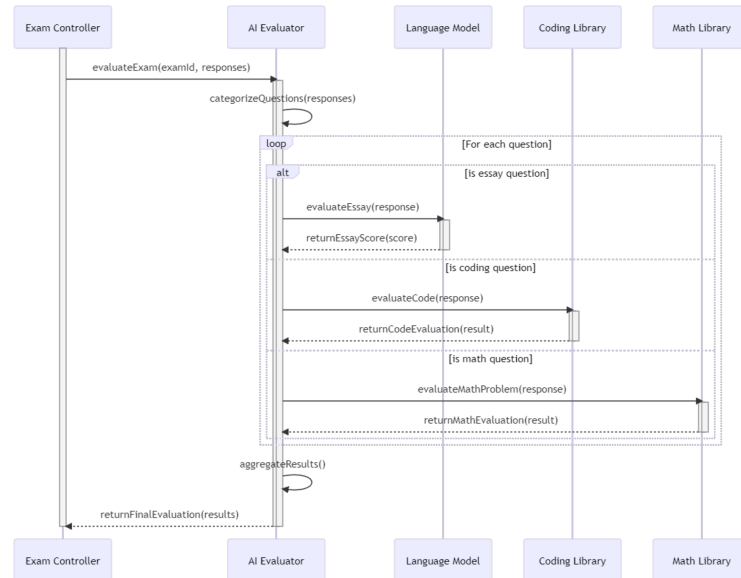


Figure 3.7: AI Evaluation Sequence Diagram

3.3 Data Design

Entities:

- User: Stores user information (teachers and students)
- Course: Represents courses offered
- Exam: Contains exam details
- Question: Stores individual exam questions
- Response: Records student answers to questions
- Result: Stores overall exam results for students

Key Relationships:

- Users create and take exams
- Courses contain exams
- Exams include questions
- Users submit responses to questions

- Exams generate results for users

Storage:

- Relational database (e.g., PostgreSQL or MySQL)
- Each entity corresponds to a database table
- Foreign keys implement relationships between tables

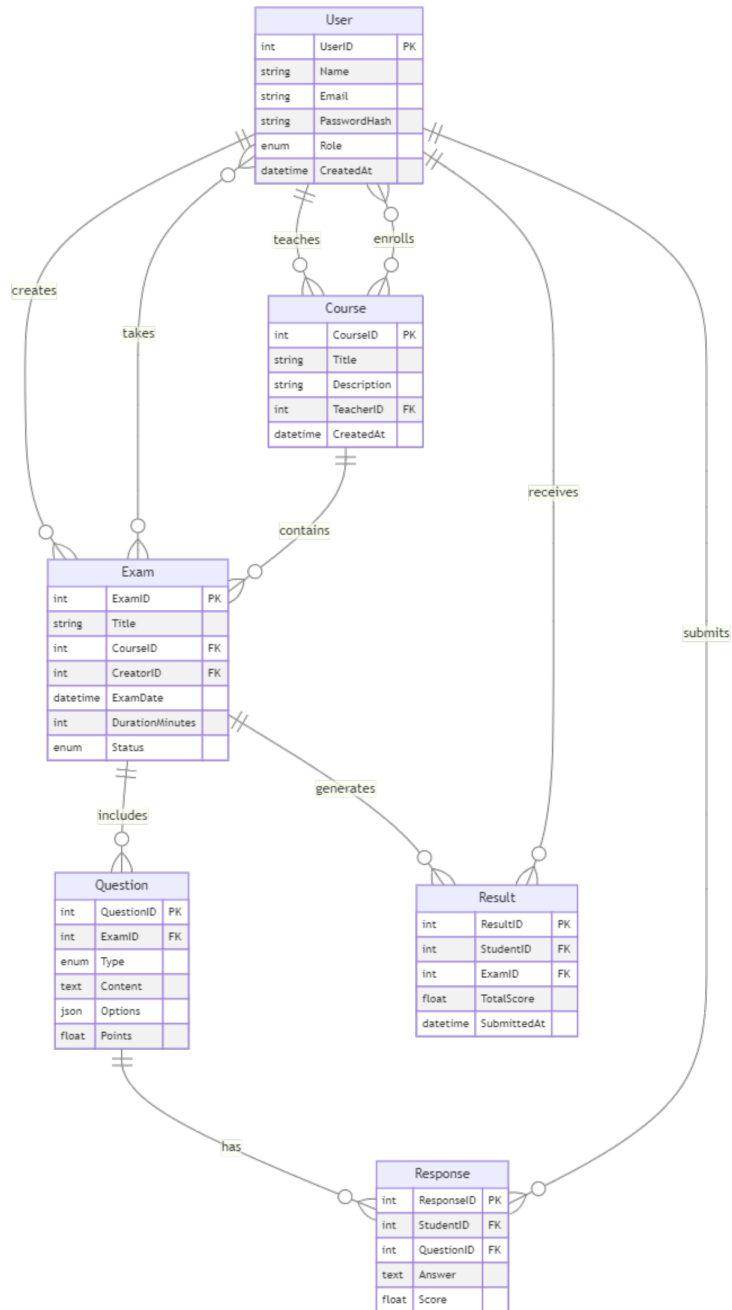


Figure 3.8: ERD

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